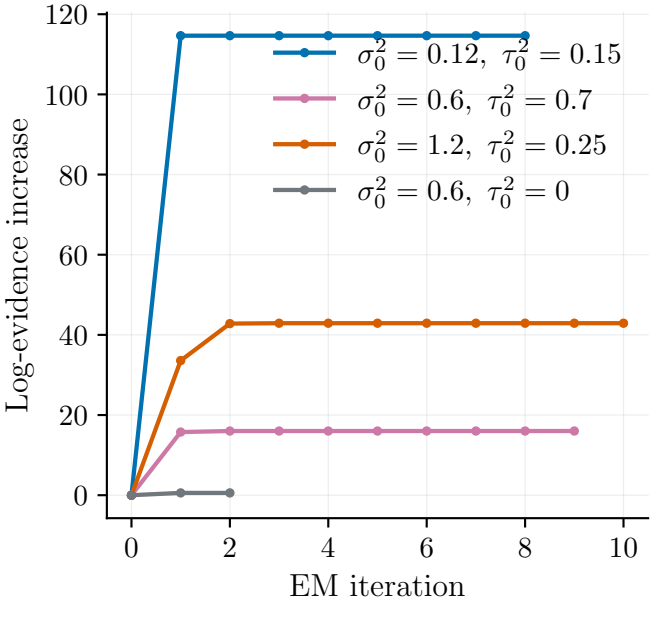
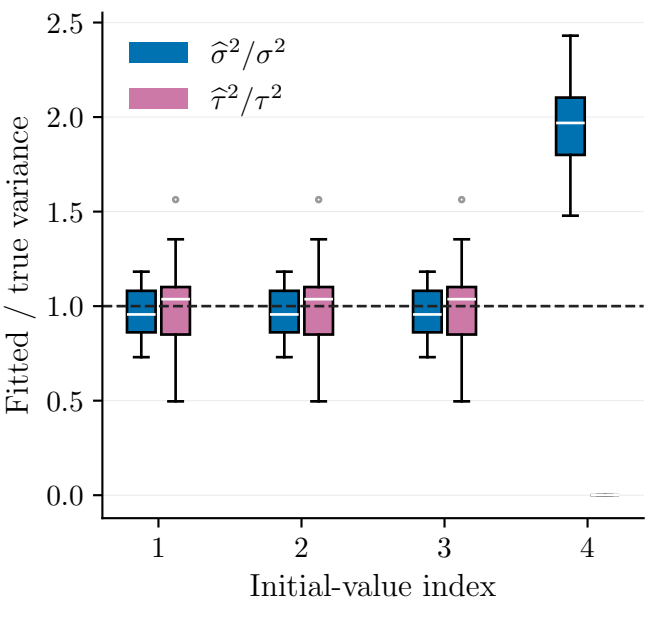
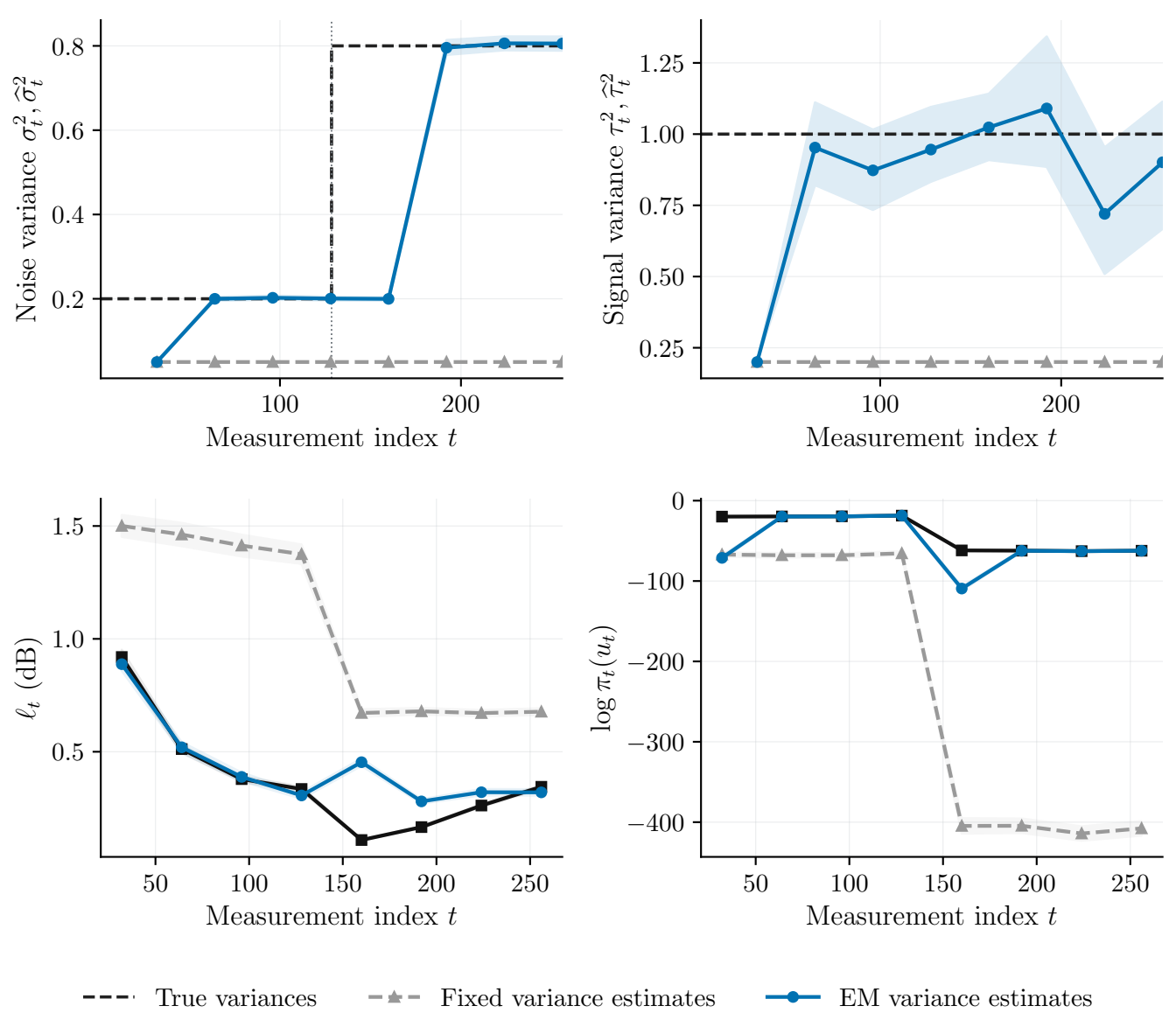
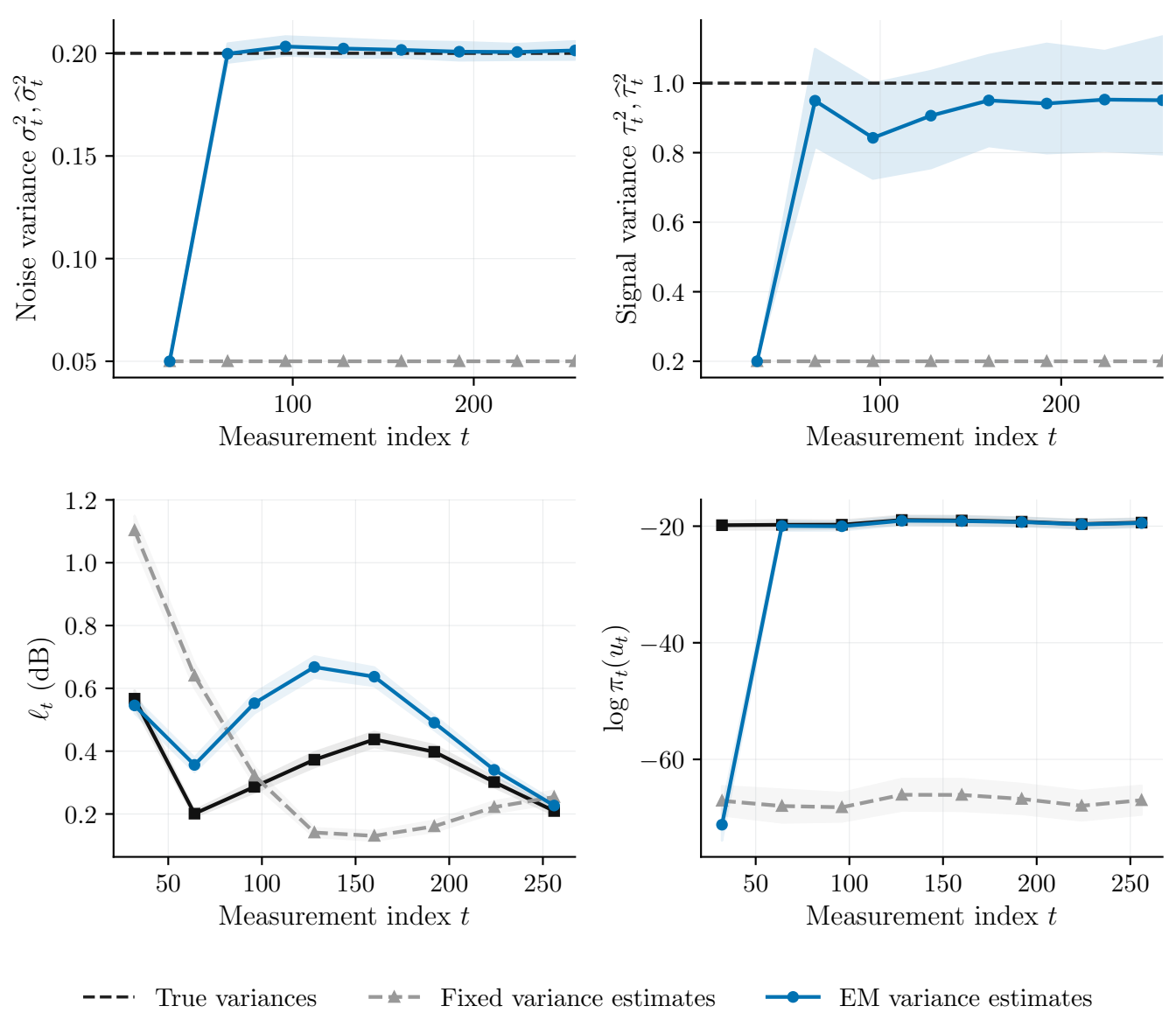
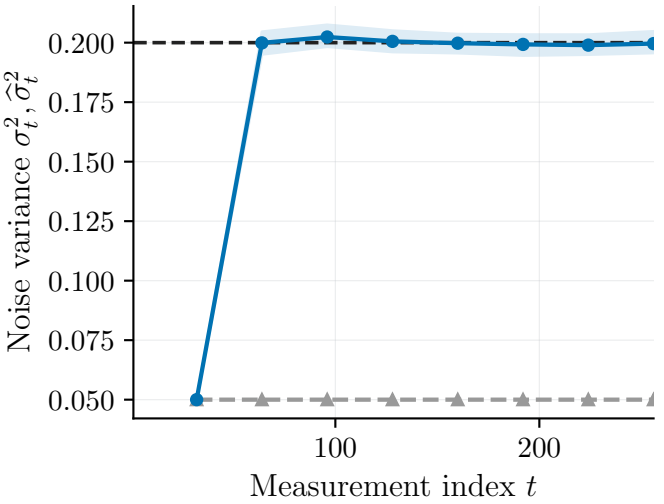
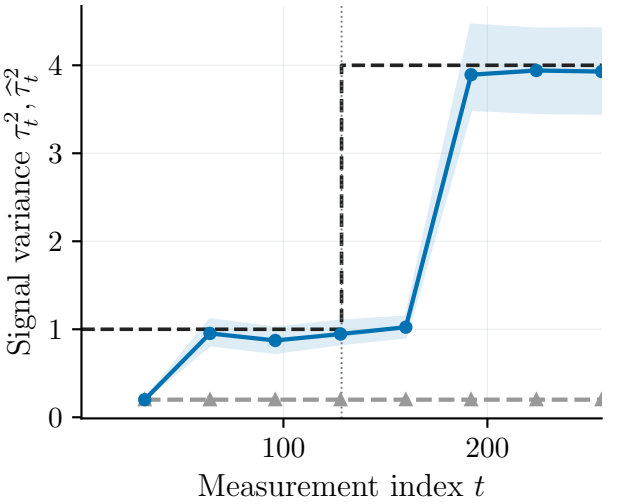

Log-evidence increase
EM iteration
$\sigma_0^2 = 0.12,\ \tau_0^2 = 0.15$
$\sigma_0^2 = 0.6,\ \tau_0^2 = 0.7$
$\sigma_0^2 = 1.2,\ \tau_0^2 = 0.25$
$\sigma_0^2 = 0.6,\ \tau_0^2 = 0$

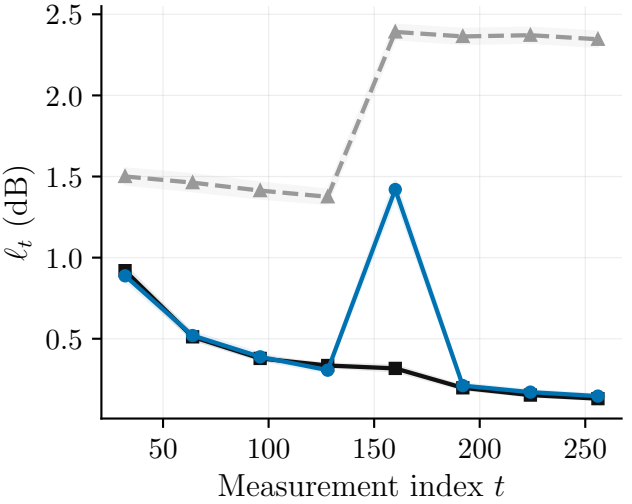

Fitted / true variance
Initial-value index
$\widehat{\sigma}^2/\sigma^2$
$\widehat{\tau}^2/\tau^2$

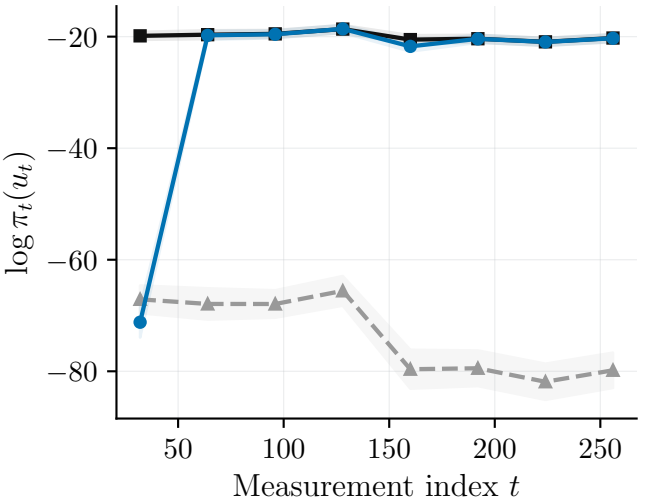

Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
0.8
0.6
0.4
0.2
100
200
Measurement index t
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
1.25
1.00
0.75
0.50
0.25
100
200
Measurement index t
ℓ_t (dB)
1.5
1.0
0.5
50
100
150
200
250
Measurement index t
$\log \pi_t(u_t)$
0
−100
−200
−300
−400
50
100
150
200
250
Measurement index t
True variances
Fixed variance estimates
EM variance estimates

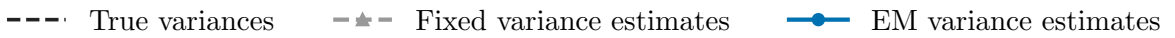

Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
0.20
0.15
0.10
0.05
100
200
Measurement index t
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
1.0
0.8
0.6
0.4
0.2
100
200
Measurement index t
ℓ_t (dB)
1.2
1.0
0.8
0.6
0.4
0.2
50
100
150
200
250
Measurement index t
$\log \pi_t(u_t)$
−20
−40
−60
50
100
150
200
250
Measurement index t
True variances
Fixed variance estimates
EM variance estimates

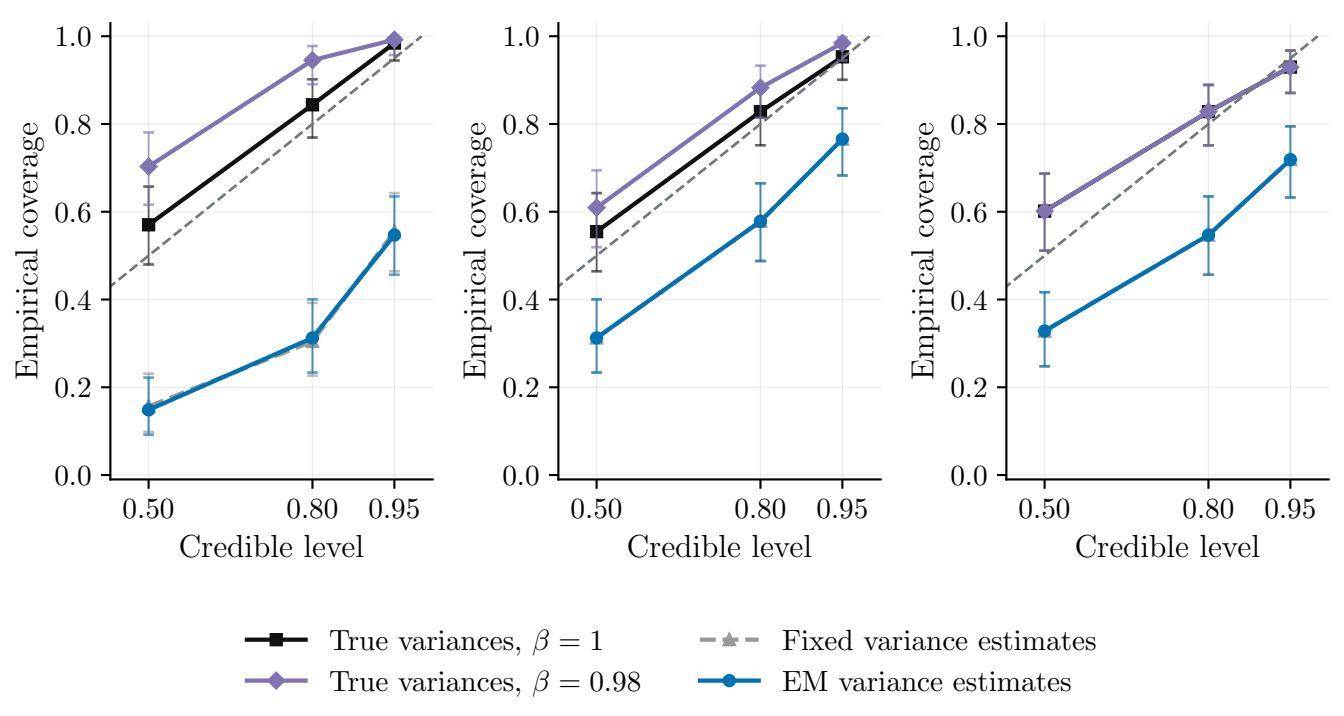

0.200
0.175
0.150
0.125
0.100
0.075
0.050
Noise variance $\sigma_t^2, \widehat{\sigma}_t^2$
100
200
Measurement index t

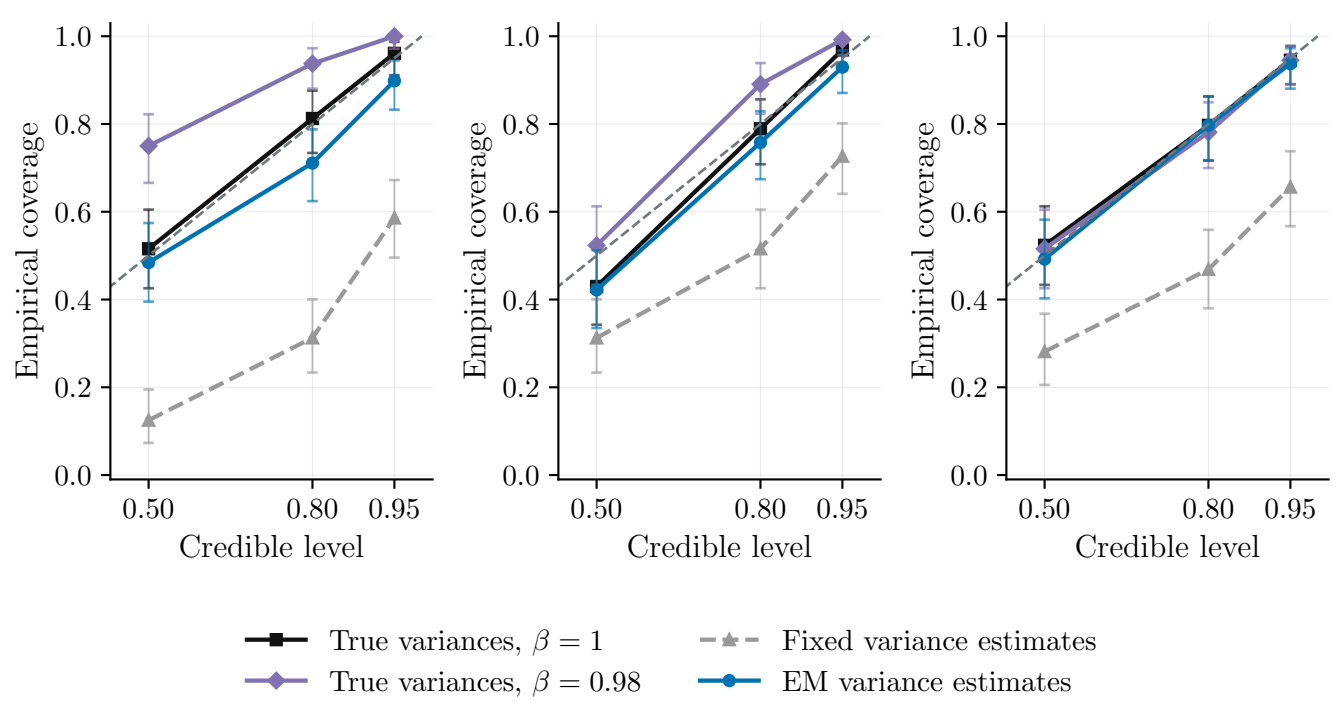

4
3
2
1
0
Signal variance $\tau_t^2, \widehat{\tau}_t^2$
100
200
Measurement index t

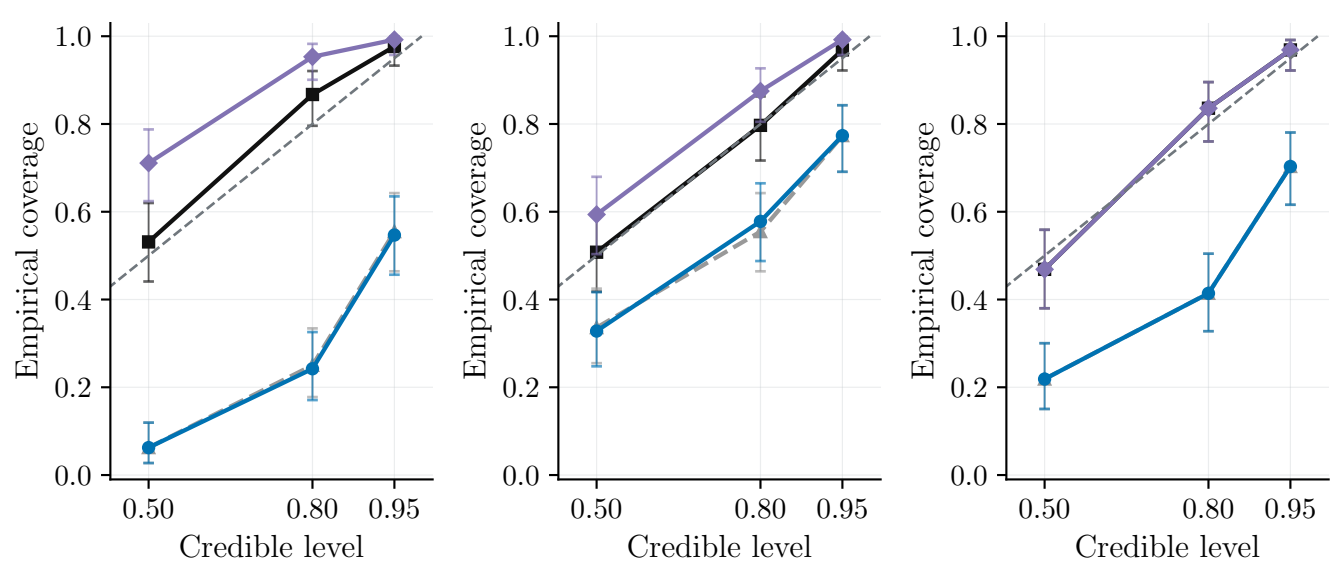

2.5
2.0
1.5
1.0
0.5
ℓ_t (dB)
50
100
150
200
250
Measurement index t

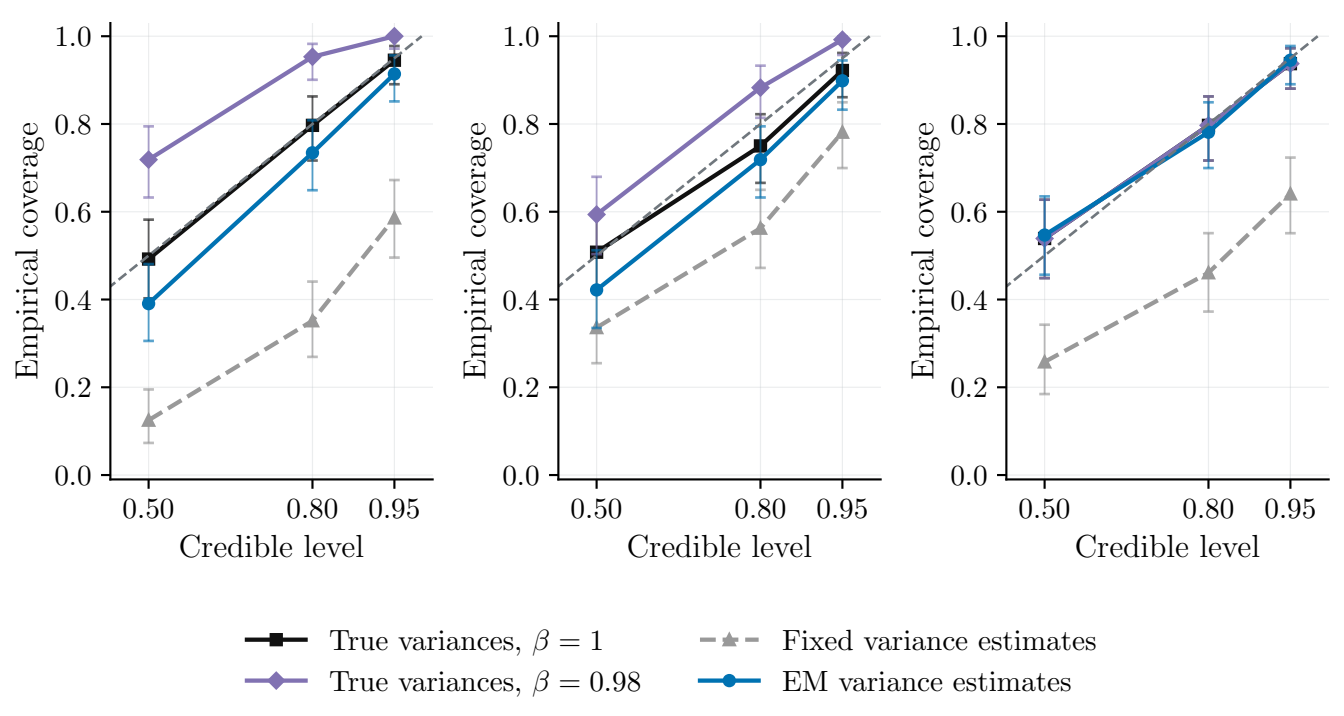

−20
−40
−60
−80
$\log \pi_t(u_t)$
50
100
150
200
250
Measurement index t

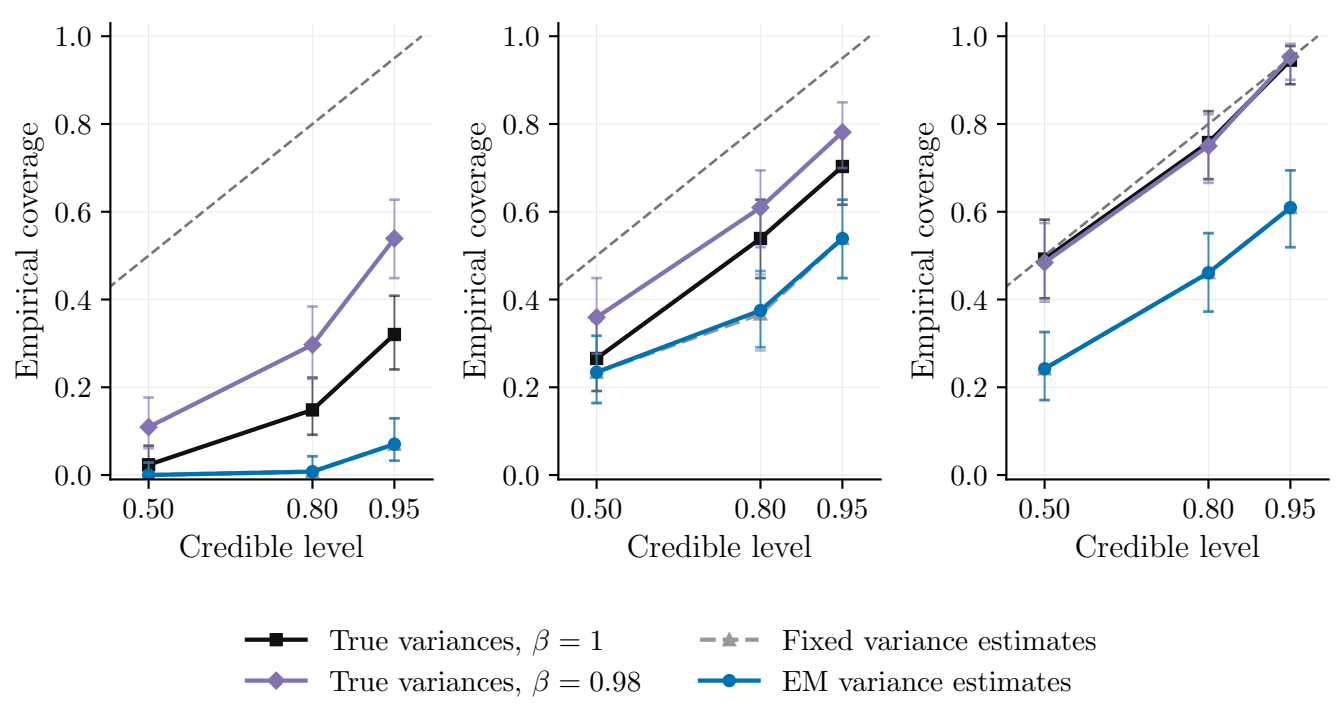

True variances
Fixed variance estimates
EM variance estimates

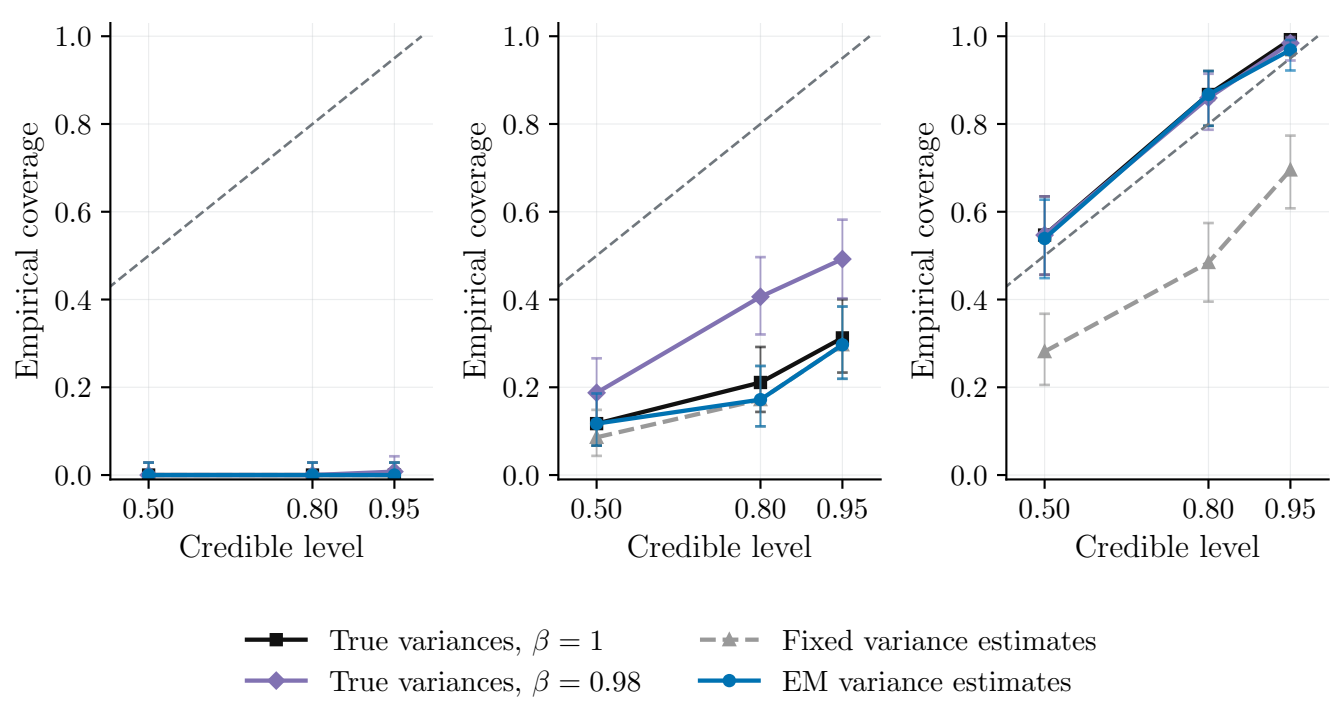

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

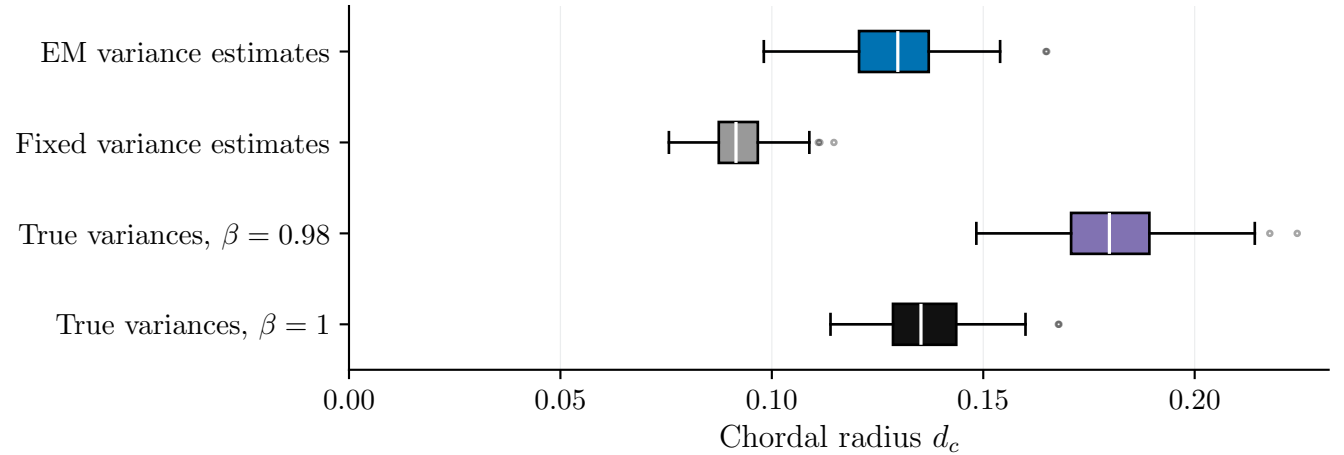

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

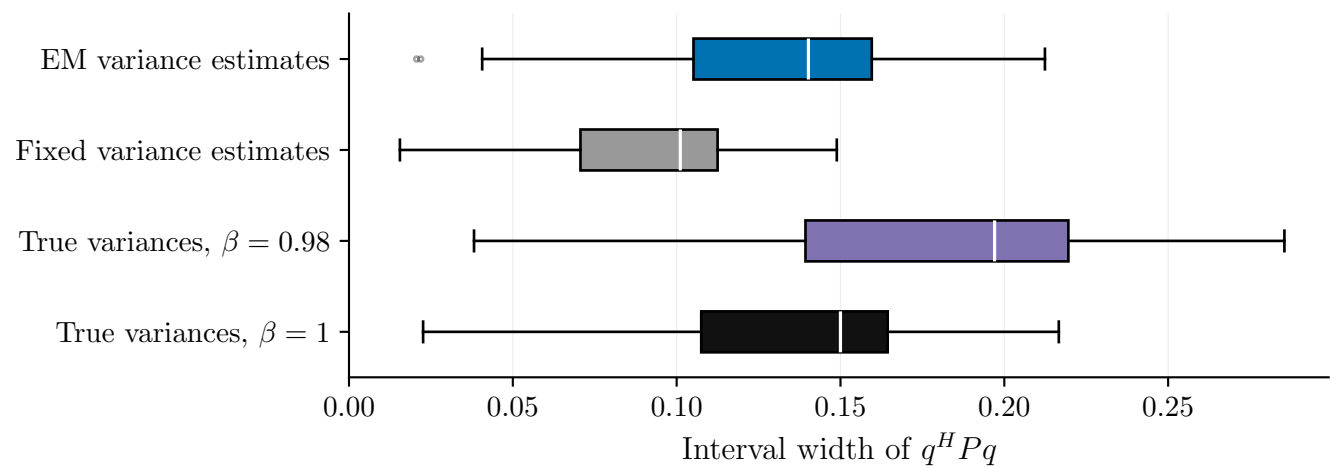

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

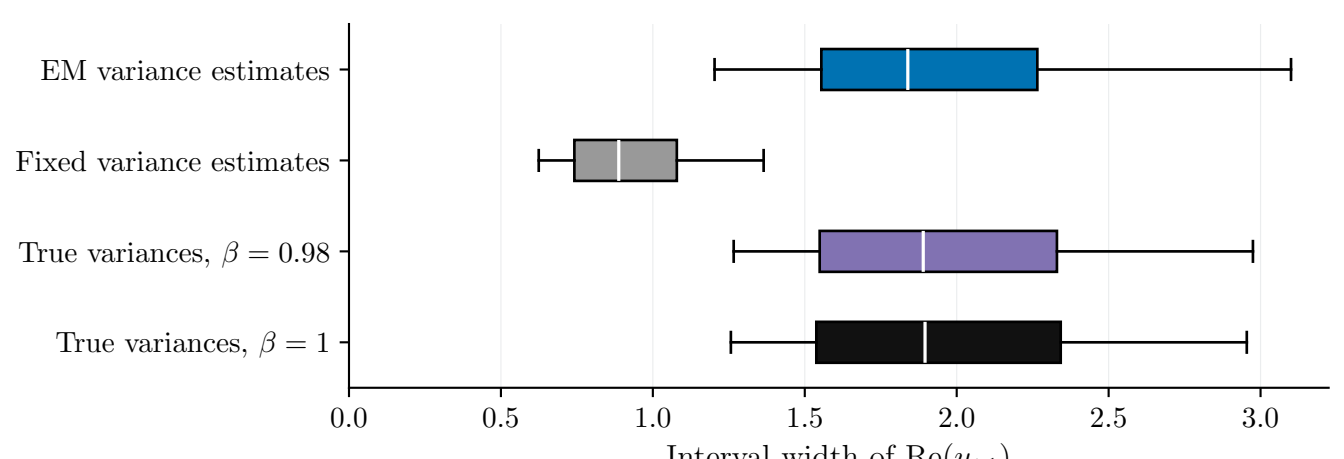

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

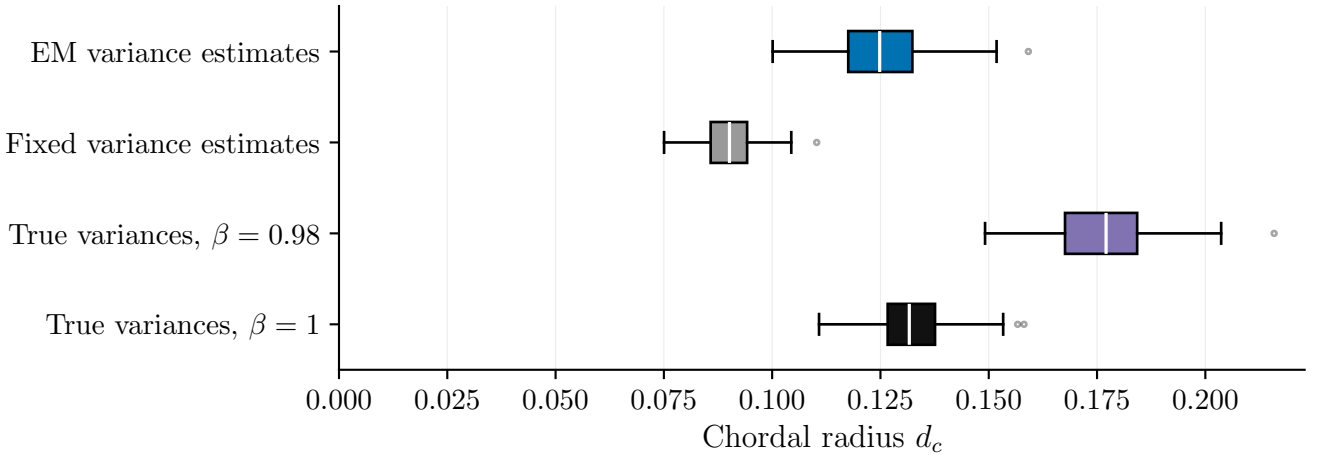

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

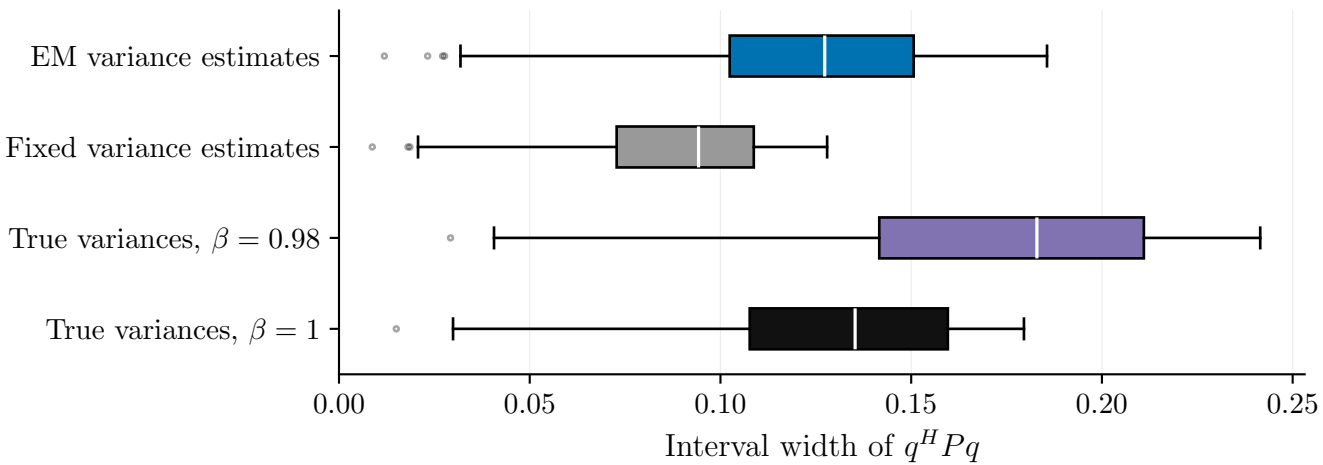

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.98$
Fixed variance estimates
EM variance estimates

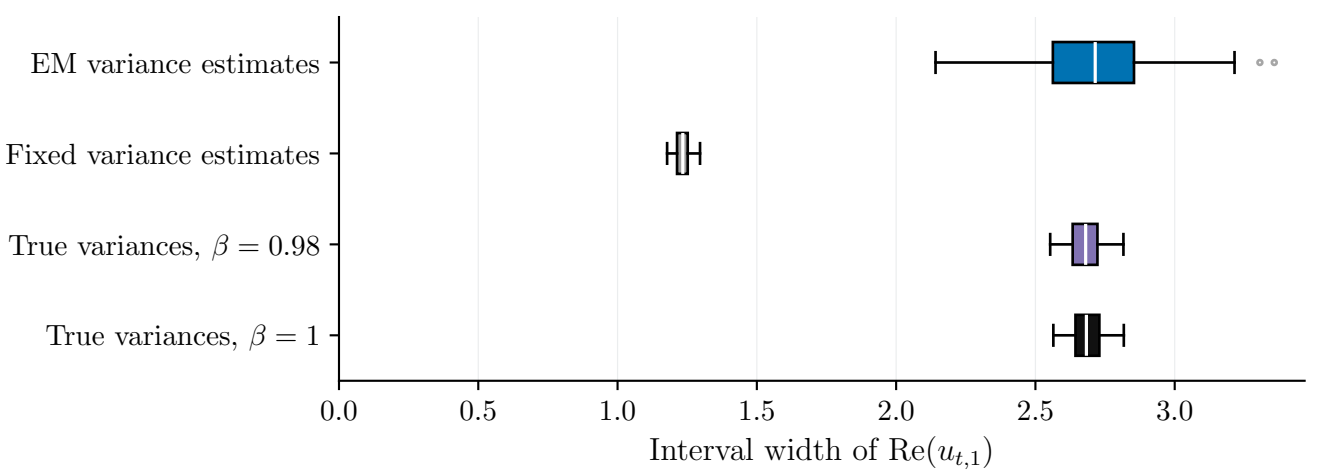

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
Chordal radius d_c

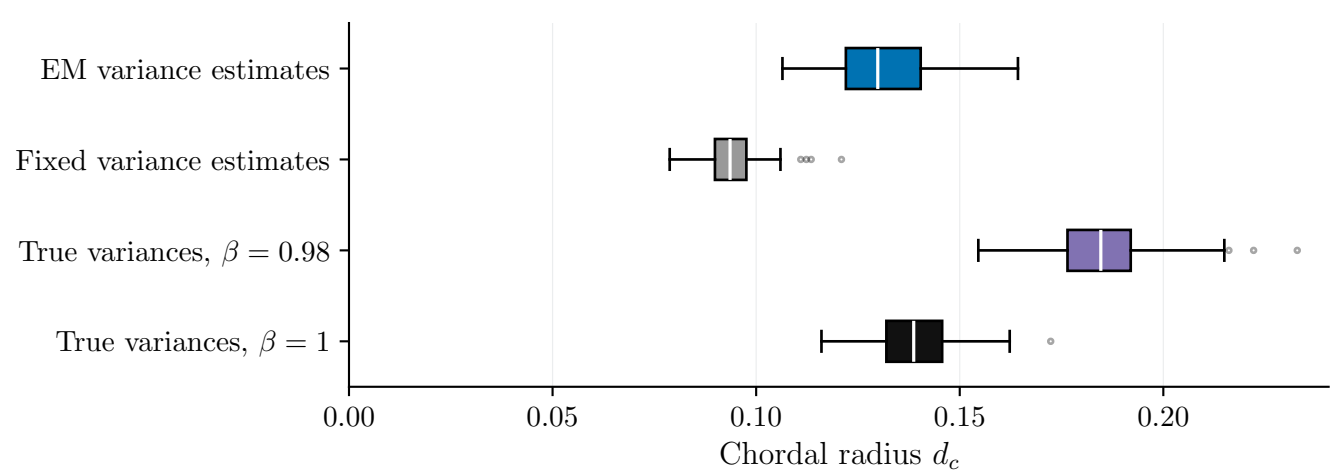

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H P q$

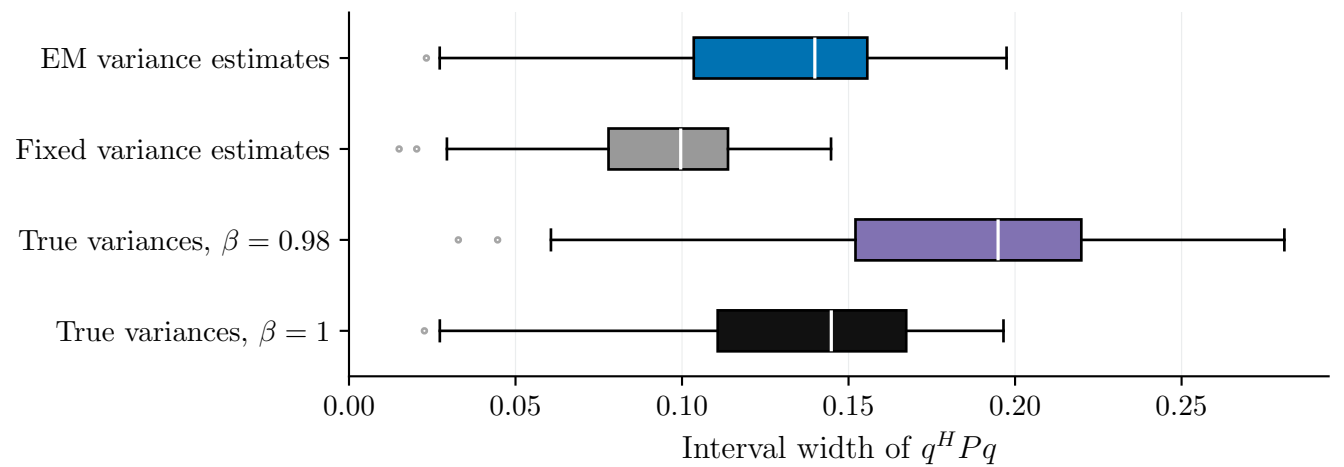

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

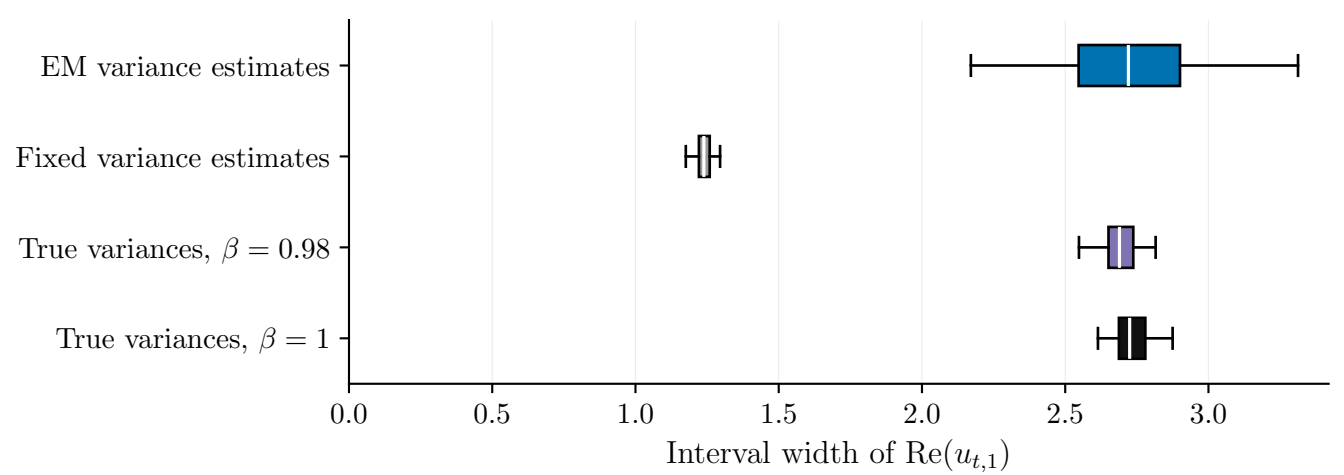

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Chordal radius d_c

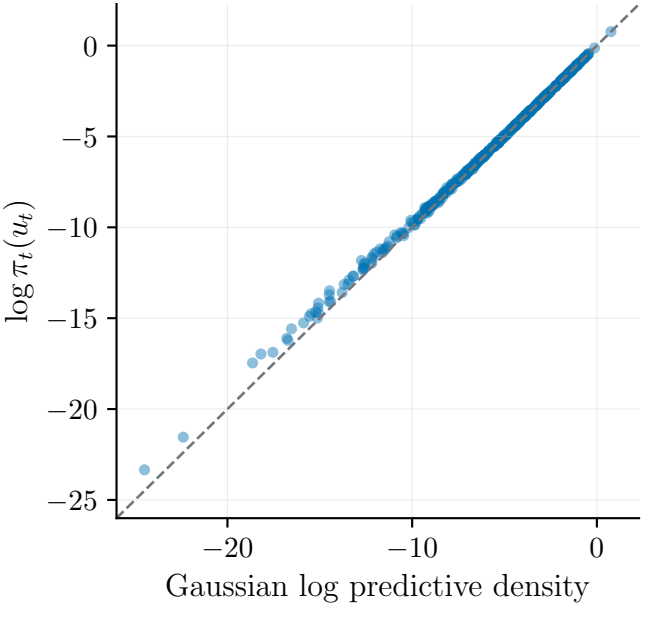

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H P q$

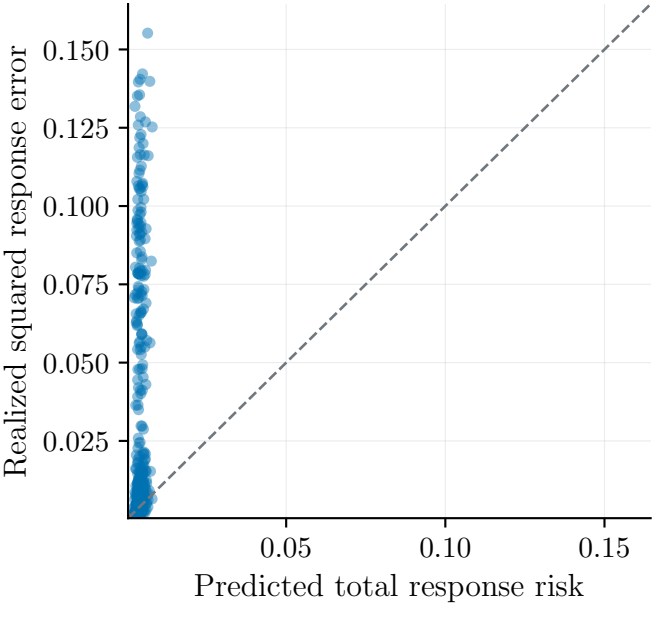

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

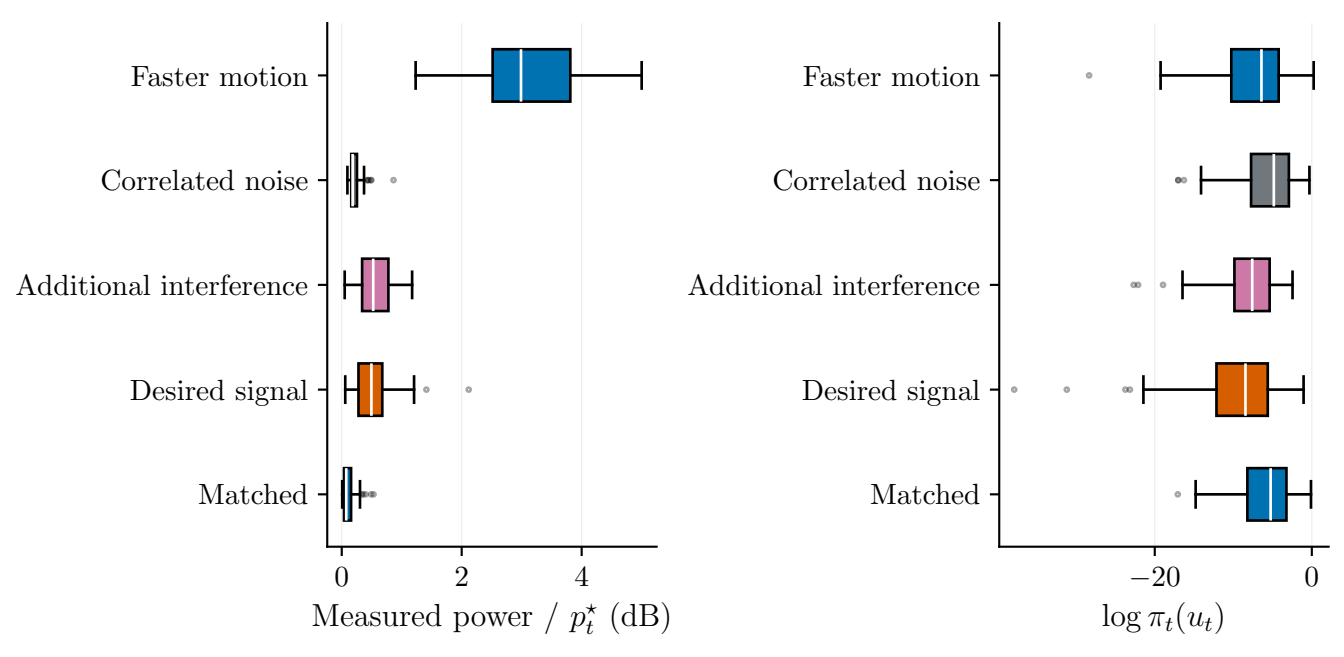

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
Chordal radius d_c

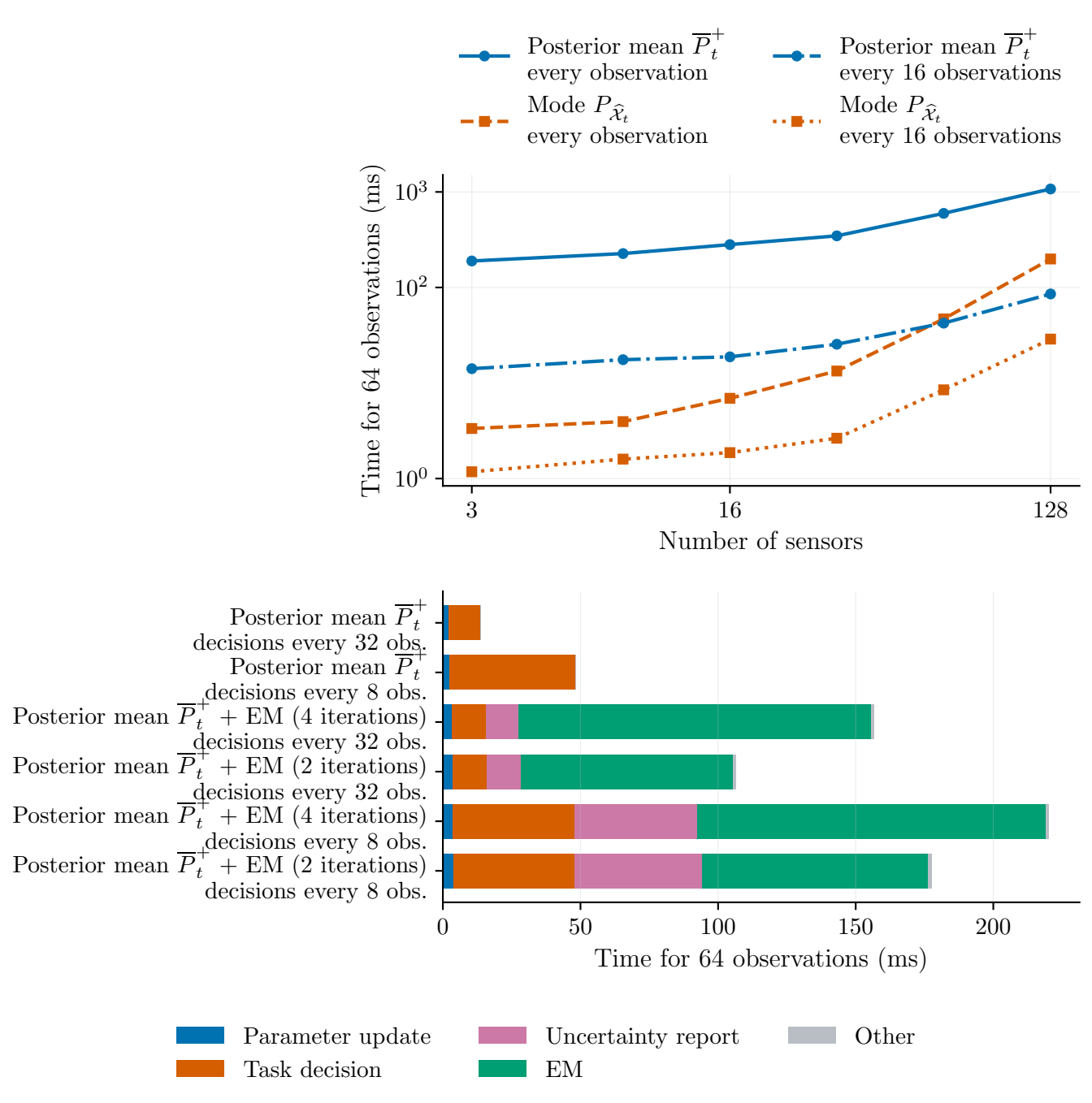

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H P q$

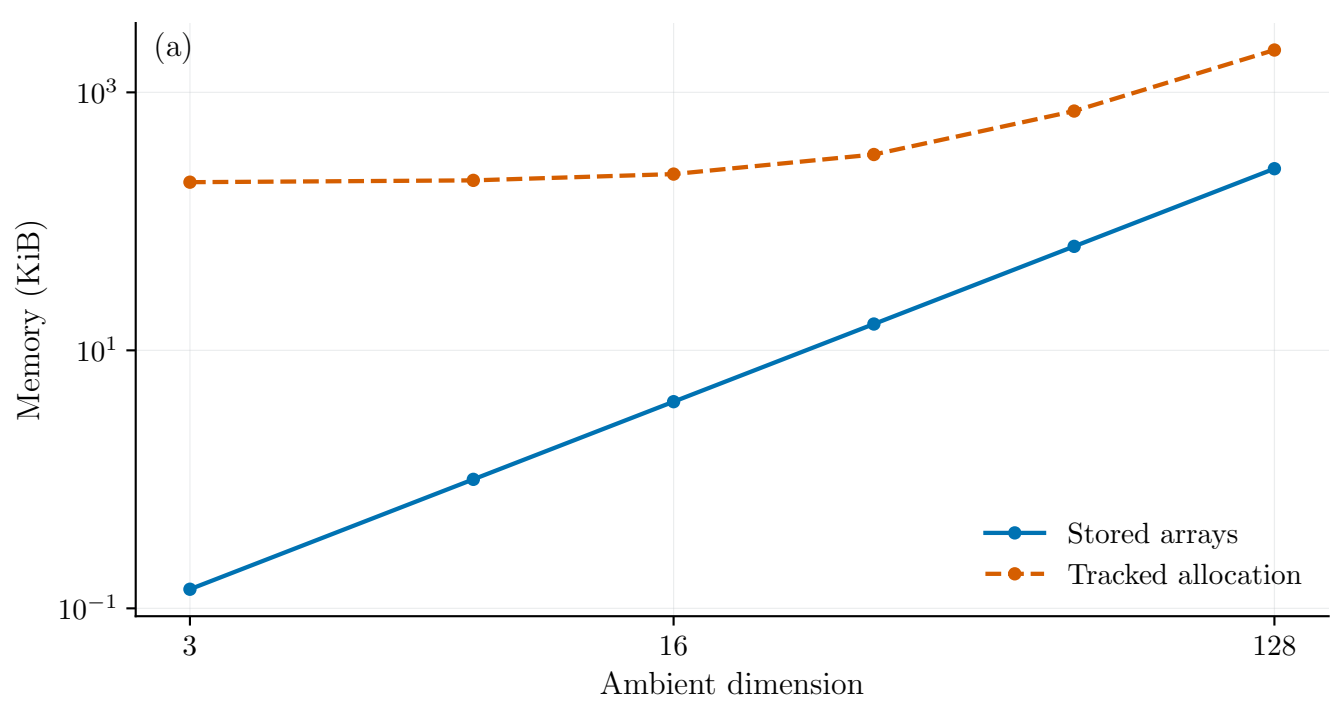

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.98$
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

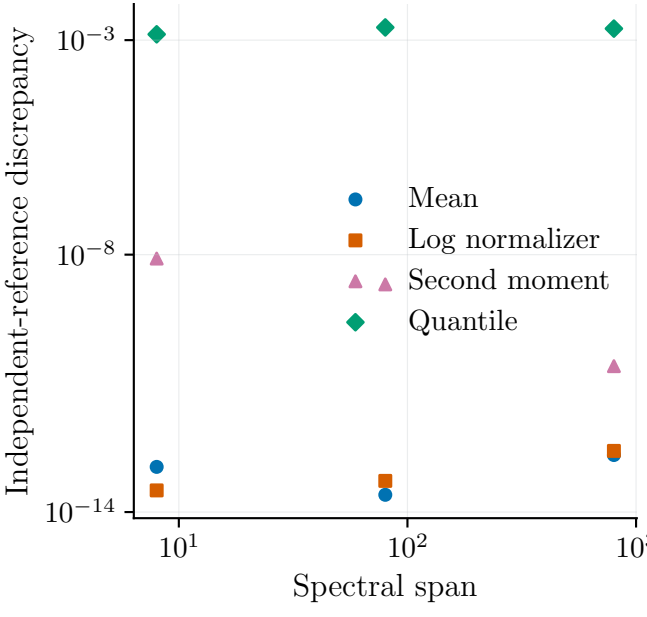

0
−5
−10
−15
−20
−25
$\log \pi_t(u_t)$
−20
−10
0
Gaussian log predictive density

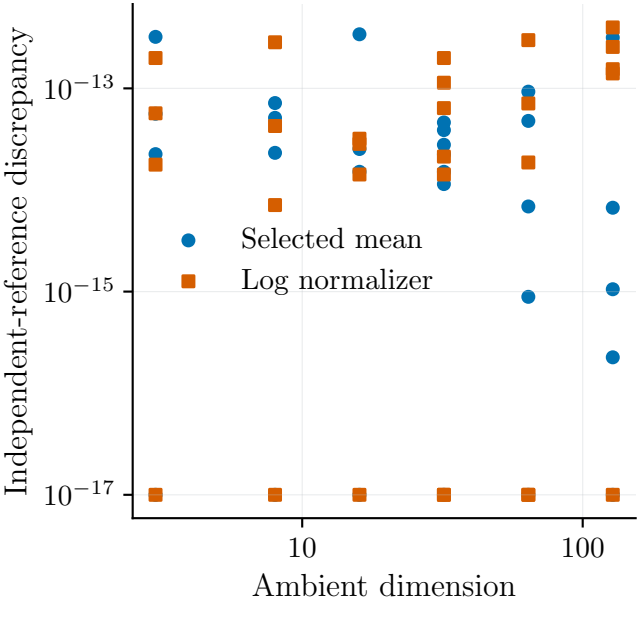

Realized squared response error
0.150
0.125
0.100
0.075
0.050
0.025
0.05
0.10
0.15
Predicted total response risk

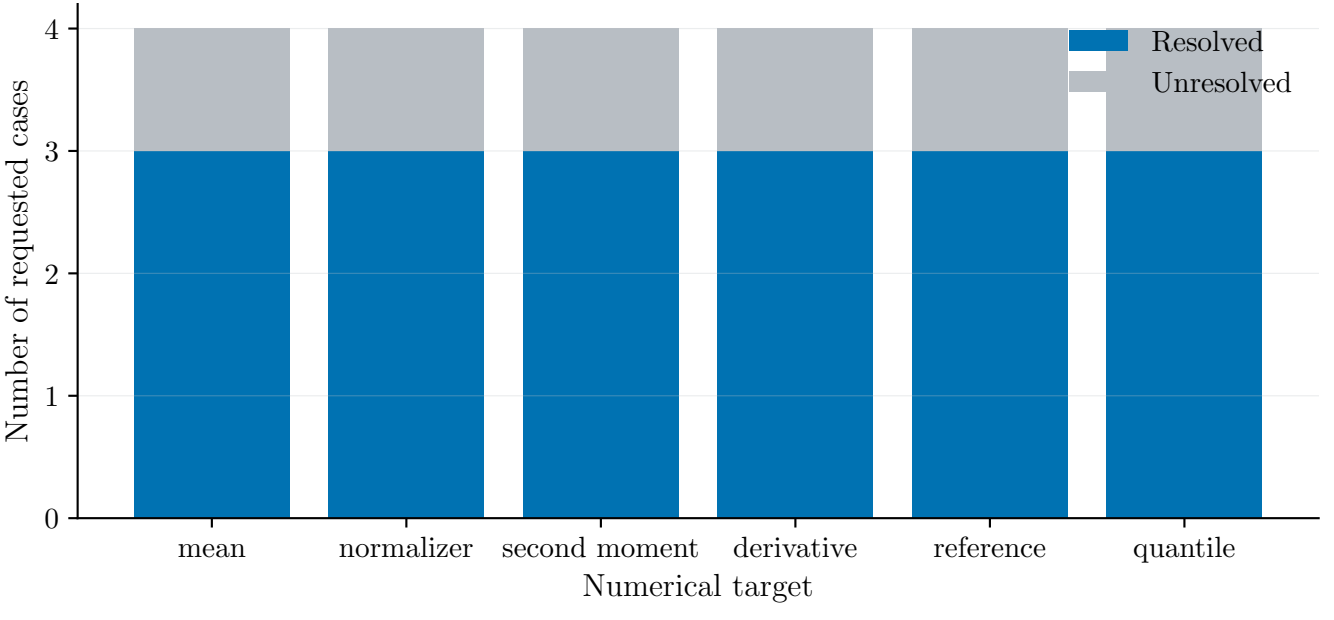

Faster motion
Correlated noise
Additional interference
Desired signal
Matched
0
2
4
Measured power / $p_t^\star$ (dB)
Faster motion
Correlated noise
Additional interference
Desired signal
Matched
−20
0
$\log \pi_t(u_t)$

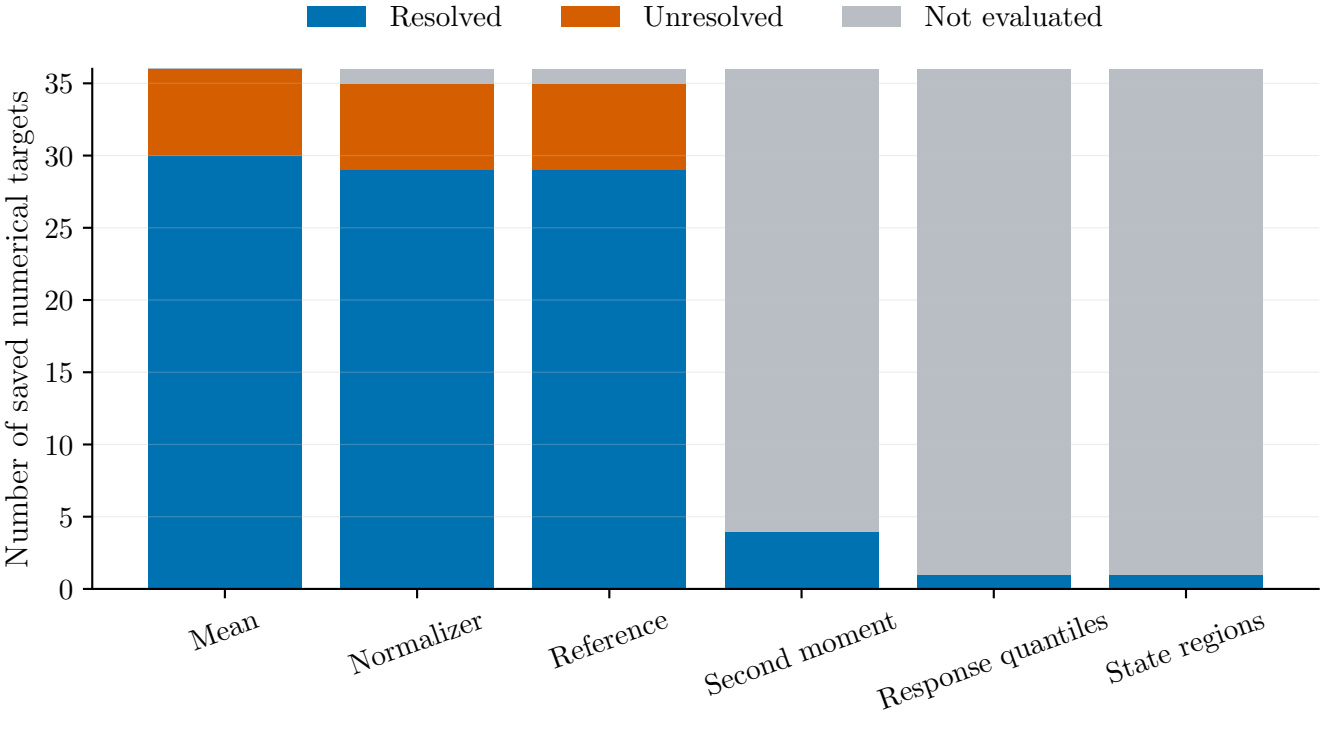

Posterior mean $\overline{P}_t^+$ every observation
Posterior mean $\overline{P}_t^+$ every 16 observations
Mode $P_{\widehat{\mathcal{X}}_t}$ every observation
Mode $P_{\widehat{\mathcal{X}}_t}$ every 16 observations
Time for 64 observations (ms)
10^3
10^2
10^0
3
16
128
Number of sensors
Posterior mean $\overline{P}_t^+$ decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ decisions every 8 obs.
Posterior mean $\overline{P}_t^+$ + EM (4 iterations) decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ + EM (2 iterations) decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ + EM (4 iterations) decisions every 8 obs.
Posterior mean $\overline{P}_t^+$ + EM (2 iterations) decisions every 8 obs.
0
50
100
150
200
Time for 64 observations (ms)
Parameter update
Task decision
Uncertainty report
EM
Other


(a)
10^3
10^1
10^{-1}
Memory (KiB)
Stored arrays
Tracked allocation
3
16
128
Ambient dimension


Independent-reference discrepancy
10^{-3}
10^{-8}
10^{-14}
10^{1}
10^{2}
10^{3}
Spectral span
Mean
Log normalizer
Second moment
Quantile


Independent-reference discrepancy
10^{-13}
10^{-15}
10^{-17}
10
100
Ambient dimension
Selected mean
Log normalizer


Resolved
Unresolved
Number of requested cases
0
1
2
3
4
mean
normalizer
second moment
derivative
reference
quantile
Numerical target


Resolved
Unresolved
Not evaluated
Number of saved numerical targets
0
5
10
15
20
25
30
35
Mean
Normalizer
Reference
Second moment
Response quantiles
State regions